CMSE 381 · Section 001

Student Exam Score Prediction

Predicting Student Exam Scores Using Machine Learning

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Background and Motivation

- Dataset: 6,607 students, 19 predictors, 1 target (Exam Score)
- Predictors cover study habits, attendance, family background, and school environment
- Preprocessing: missing values filled by mode, categorical variables encoded, features standardized with StandardScaler

Research Questions

1. Can we accurately predict student exam scores from 19 predictors, and which ones matter most?

2. Can we classify students into Low, Medium, and High performance groups based on these predictors?

3. Do Lasso and Ridge regression outperform OLS and effectively produce automatic feature selection?

4. Can we compress the predictor space into fewer principal components without losing predictive power, and what do those components represent?

Methods Overview

Classification

Logistic Regression (baseline) vs Random Forest + GridSearchCV
10-fold Stratified CV · 80/20 train-test split

OLS Regression

Multiple linear regression on all 19 predictors
10-fold CV for R^2 and RMSE · Coefficient interpretation

Ridge & Lasso

CV-selected alpha via RidgeCV / LassoCV
Lasso provides automatic feature selection

PCA + PCR

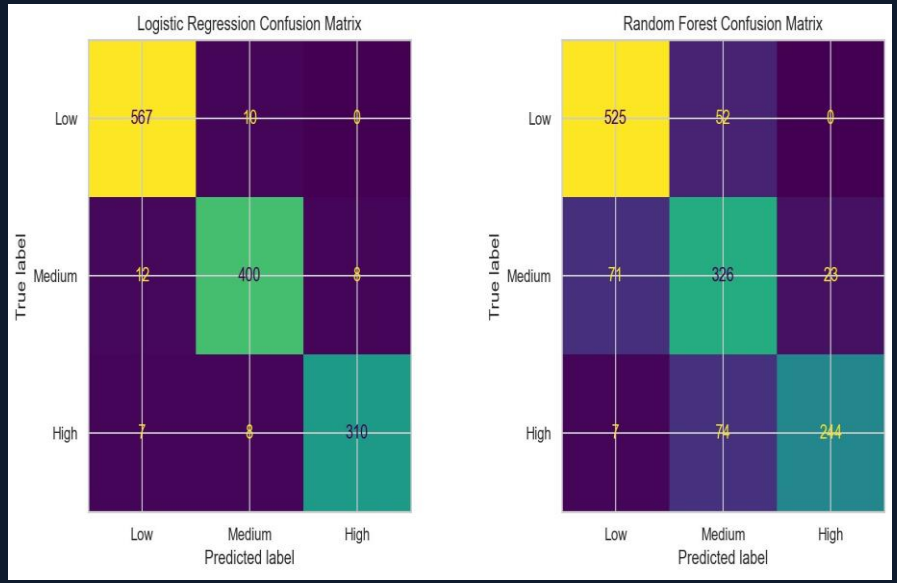
Scree plot + cumulative variance analysis
CV over 1–19 components · PC1 loading interpretation

Logistic Regression

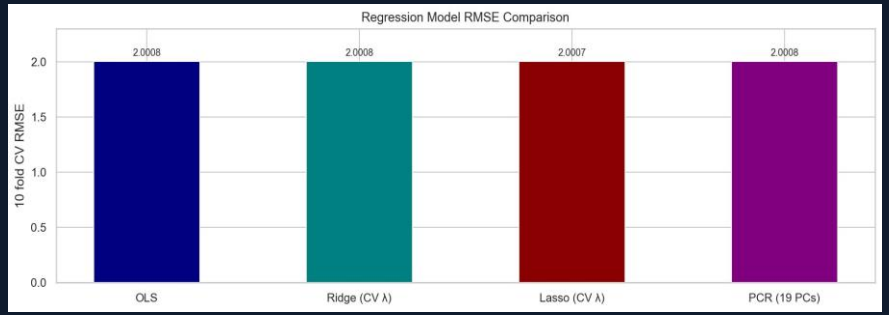
Outperforms Random Forest by ~14 percentage points in test accuracy

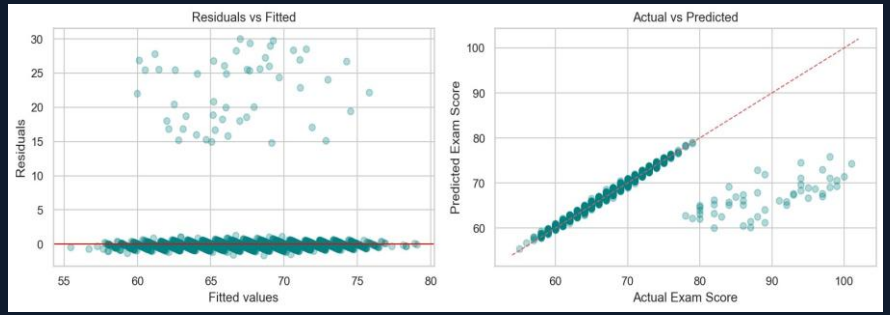
Model	CV Acc	Test Acc	Macro F1
Logistic Regression	**96.7%**	**96.6%**	**0.965**
Random Forest	83.1%	82.8%	0.821

- CV accuracy ≈ test accuracy (96.7%) → generalizes well
- Low group recall = 98% → strong at identifying at-risk students
- Logistic Regression outperforms the more complex Random Forest

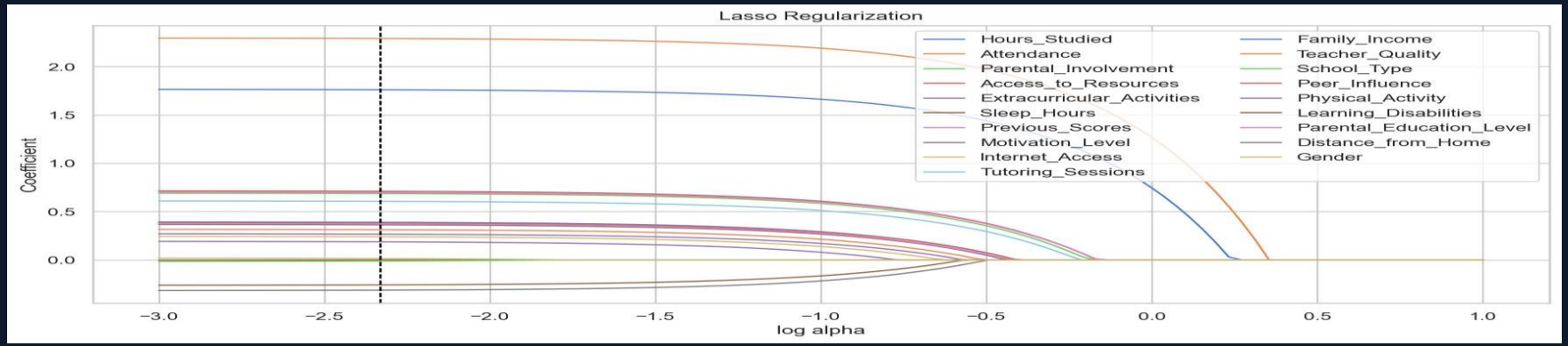


Model Comparison: OLS, Ridge, Lasso, PCR

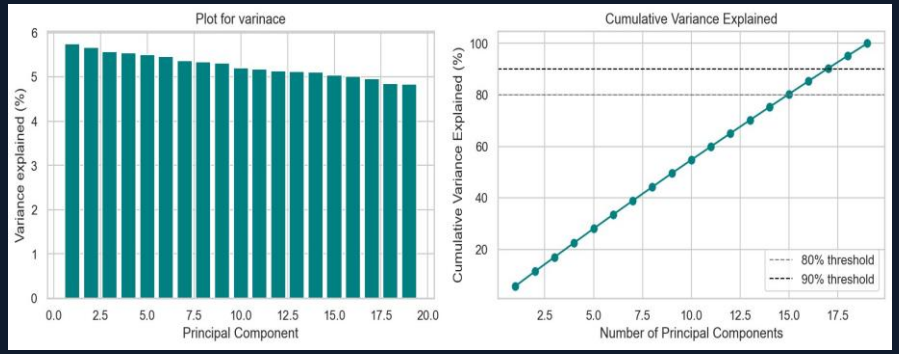


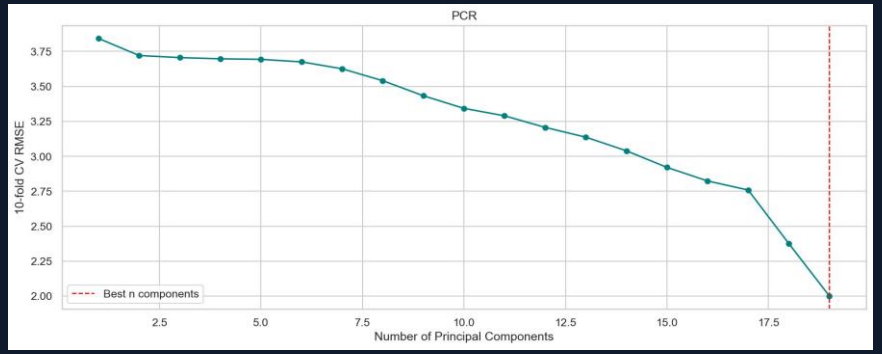


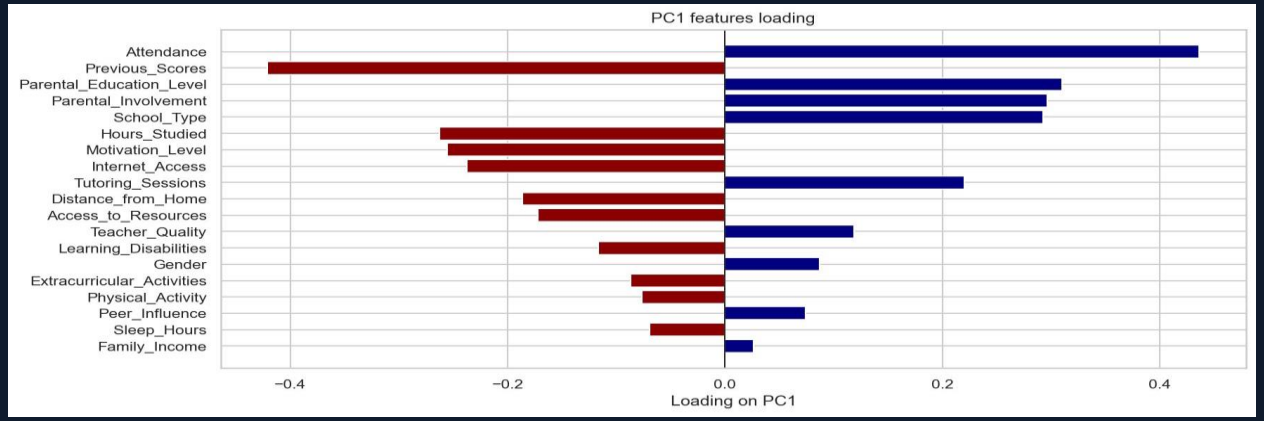
All 4 models: CV R2 ~ 0.732, RMSE ~ 2.00 | Attendance +2.3 pts/SD Hours Studied +1.7 pts/SD | Lasso zeroed only Sleep_Hours



Features from PCA and PCR







- No single PC dominates — all components contribute roughly equally
- CV selected all 19 PCs → no compression benefit for prediction
- PC1 = student engagement axis: Attendance, Study Hours, Motivation, Tutoring
- High PC1 score → more engaged student → higher exam score

06 CONCLUSION

Summary

✓ 96.6% classification accuracy — Logistic Regression outperformed Random Forest by ~14%

✓ $R^2 = 0.732$, RMSE ≈ 2 pts across all regression models — regularization gave no gain

✓ Attendance & Hours Studied are the strongest predictors across all methods

✓ PCA: 19 predictors are not redundant; PC1 captures the student engagement dimension

Next steps: SVMs, interaction terms, gradient boosting